

Academic & Conference Research Compendium: Ring -3

Hacking Conferences, Academic Papers, Industry Research, and Community
Tools

TLP:WHITE — PUBLIC RELEASE

RingWatch Research Division
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Academic & Conference Research Compendium: Ring -3 (Intel ME/CSME) Attacks and Defenses

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Scope: Comprehensive catalog of academic papers, hacking conference presentations, red team forum tools, and reputable security research concerning Ring -3 (Intel Management Engine / Converged Security and Management Engine) attack and defense.

1. Hacking Conference Presentations

1.1 Black Hat USA

Black Hat USA 2017 — "Intel AMT: Stealth Breakthrough"

- **Authors:** Dmitry Evdokimov (@evdokimovds), Alexander Ermolov, Maksim Malyutin (Embedi)
- **URL:** <https://blackhat.com/docs/us-17/thursday/us-17-Evdokimov-Intel-AMT-Stealth-Breakthrough.pdf>
- **White Paper:** <https://blackhat.com/docs/us-17/thursday/us-17-Evdokimov-Intel-AMT-Stealth-Breakthrough-wp.pdf>
- **Summary:** Detailed analysis of Intel AMT vulnerabilities post-CVE-2017-5689. Demonstrated new exploitation vectors for AMT, including stealth persistence mechanisms. Showed how AMT can be used as a covert network channel independent of the host OS. Presented at Black Hat USA 2017 and later updated at GSEC/HITB 2017.
- **Key Contributions:**
 - New AMT exploitation vectors beyond the original authentication bypass
 - Stealth network communication via AMT's independent network stack
 - Persistence mechanisms that survive OS reinstalls
 - Practical exploitation scenarios for enterprise environments

Black Hat USA 2017 — Intel-SA-00086 Disclosure

- **Authors:** Mark Ermolov (@markel_), Maxim Goryachy (@h0tmax) — Positive Technologies
- **Summary:** Disclosure of the now-infamous SA-00086 vulnerability affecting Intel ME 6.x through 11.x. Demonstrated JTAG activation via USB 3.0 debug port, giving full system access through the ME. This was the talk that put Ring -3 security on the map for the broader security community.
- **Key Contributions:**
 - First public demonstration of ME JTAG activation
 - Full system compromise via USB 3.0 debug port
 - Proof that Ring -3 is accessible to attackers with physical access
 - Released as IntelTXE-PoC (545 GitHub stars)

Black Hat USA 2019 — "Behind the Scenes of Intel Security and Manageability Engine"

- **Authors:** Shai Hasarfaty (Principal Security Research Engineer, Intel), Yanai Moyal (Security Researcher, Intel)

- **URL:** <https://papers.put.as/papers/firmware/2019/us-19-Hasarfaty-Behind-The-Scenes-Of-Intel-Security-And-Manageability-Engine.pdf>
- **Summary:** Intel's own presentation of ME security architecture. Covered ME threat model, security boundaries, and internal defense mechanisms. Rare public insight from Intel itself about ME internals.
- **Key Contributions:**
 - Intel's official ME threat model
 - Description of ME security boundaries and isolation mechanisms
 - Internal defense mechanisms within the ME
 - Intel's perspective on ME vulnerability disclosure

1.2 DEF CON

DEF CON 26 (2018) — "The Ring 0 Facade: Awakening the Processor's Inner Demons"

- **Author:** Christopher Domas (@xoreaxeaxeax)
- **URL:** <https://infocon.org/cons/DEF%20CON/DEF%20CON%2026/DEF%20CON%2026%20presentations/Christopher%20Domas/DEFCON-26-Christopher-Domas-The-Ring-0-Facade-Updated.pdf>
- **Summary:** While focused on Ring 0 (kernel) → below Ring 0 escalation via processor microcode, this talk is foundational for Ring -3 research because it demonstrated that the traditional ring hierarchy is a "facade" — there are undocumented privilege levels below Ring 0 that can be reached through processor features.
- **Key Contributions:**
 - Demonstrated that processor privilege rings are not as well-isolated as assumed
 - Showed microcode-based escalation below Ring 0
 - Conceptual framework for understanding Ring -1/-2/-3 as reachable targets
 - Influenced subsequent Ring -3 research by demonstrating the ring hierarchy is penetrable

1.3 Chaos Communication Congress (CCC)

30C3 (2013) — "Persistent, Stealthy, Remote-controlled Dedicated Hardware Malware"

- **URL:** <https://infocondb.org/con/ccc/30c3/persistent-stealthy-remote-controlled-dedicated-hardware-malware>
- **Summary:** Foundational talk about hardware-based malware persistence. Predates the ME-specific research but established the concept of dedicated hardware malware that persists across reboots and OS reinstalls. This is the conceptual ancestor of Ring -3 rootkit research.
- **Key Contributions:**
 - First public discussion of hardware-level malware persistence
 - Conceptual framework for "stealthy, remote-controlled" hardware implants
 - Influenced later ME research by establishing the threat model

34C3 (2017) — "Inside Intel Management Engine"

- **Authors:** Maxim Goryachy and Mark Ermolov (Positive Technologies)
- **URL:** https://media.ccc.de/v/34c3-8762-insideintelmanagement_engine
- **Summary:** Deep technical dive into the SA-00086 vulnerability. Covered ME firmware internals, the JTAG activation mechanism, and demonstrated live exploitation. More technical than the Black Hat presentation, with extensive Q&A from the hacker community.
- **Key Contributions:**
 - Detailed ME firmware architecture walkthrough
 - Live demonstration of JTAG-based ME compromise

- Community Q&A revealing additional attack vectors
- Discussion of ME firmware modification for persistence

1.4 Other Conferences

GSEC/HITB 2017 — "Intel AMT" (Part 2)

- **Authors:** Evdokimov, Ermolov, Malyutin (Embedi)
- **PDF:** <https://github.com/flothrone/intelme/blob/main/IntelAMTpart2.pdf>
- **Summary:** Updated version of the Black Hat AMT talk with additional findings and exploitation scenarios.

SICHERHEIT 2017 — "LONGKIT"

- **Authors:** Julian Rauchberger, Robert Luh, Sebastian Schrittwieser (St. Poelten University of Applied Sciences, Austria)
- **PDF:** <https://www.scitepress.org/papers/2017/61656/61656.pdf>
- **Summary:** Universal framework for BIOS/UEFI rootkits in System Management Mode (SMM / Ring -2). Relevant to Ring -3 because the ME can inject SMM handlers, creating a Ring -3 → Ring -2 persistence chain.

EuroS&P 2026 — "Helltrap"

- **PDF:** <https://zxr.io/research/sion2026eurosp.pdf>
- **Summary:** Defensive approach — deliberately installing UEFI rootkit traps to catch attackers. Relevant to RingWatch because the ME could serve as an out-of-band trap deployment mechanism.

2. Academic Papers

2.1 Firmware Security & Integrity

BootKeeper (2019)

- **Citation:** Chevalier, R., Villatel, M., Plaquin, D., et al. "BootKeeper: Validating Software Integrity Properties on Boot Firmware Images." arXiv:1903.12505, 2019.
- **URL:** <https://arxiv.labs.arxiv.org/html/1903.12505>
- **Summary:** Proposes validating software integrity properties on boot firmware images by analyzing control flow and data flow. Addresses the challenge of detecting firmware modifications without a trusted anchor.
- **Relevance to RingWatch:** RingWatch's ME firmware integrity monitoring (version tracking, status register decoding) is inspired by BootKeeper's approach.

Peacock (2026)

- **Citation:** Cochavi Gorelik, H., Fadlon, O., Klimov, D., et al. "Peacock: UEFI Firmware Runtime Observability Layer for Detection and Response." arXiv:2601.07402, January 2026.
- **URL:** <https://arxiv.org/pdf/2601.07402>
- **Summary:** Proposes a runtime UEFI firmware observability layer for detection and response. Addresses the gap in runtime firmware monitoring — most firmware security tools only check at boot time, not during runtime.
- **Relevance to RingWatch:** RingWatch provides runtime ME monitoring, which is the natural extension of Peacock's approach. Peacock monitors UEFI runtime services from the OS; RingWatch monitors the ME from the OS. The ideal architecture combines both.

UEFI Vulnerability Signature Generation (2024)

- **Citation:** "UEFI Vulnerability Signature Generation using Static and Symbolic Analysis." arXiv:2407.07166v2, 2024.
- **URL:** <https://arxiv.deeppaper.ai/papers/2407.07166v2>
- **Summary:** STASE technique integrating static and symbolic analysis to automate UEFI vulnerability signature generation. Relevant for automated ME firmware vulnerability discovery.

2.2 DMA & IOMMU Security

Thunderclap (NDSS 2019)

- **Citation:** Markettos, A.T., Rothwell, C., Gutstein, B.F., et al. "Thunderclap: Exploring Vulnerabilities in Operating System IOMMU Protection via DMA from Untrustworthy Peripherals." NDSS 2019.
- **URL:** https://ndss-symposium.org/wp-content/uploads/2019/02/ndss201905A-1Markettos_paper.pdf
- **Summary:** Comprehensive analysis of IOMMU protection vulnerabilities. Demonstrated that IOMMU can be bypassed by malicious peripherals with DMA capability. Directly relevant to ME's DMA access to host memory.
- **Key Finding:** IOMMU is not a complete defense against DMA attacks — there are timing windows and configuration vulnerabilities that can be exploited.

DmaAuth (USENIX Security 2024)

- **Citation:** Wang, X. "DmaAuth: A Lightweight Pointer Integrity-based Secure Architecture to Defeat DMA Attacks." USENIX Security 2024.
- **URL:** <https://www.usenix.org/system/files/usenixsecurity24-wang-xingkai.pdf>
- **Summary:** Proposes pointer integrity as a defense against DMA attacks. More efficient than IOMMU alone.
- **Relevance to RingWatch:** RingWatch's ME DMA Protection module monitors IOMMU state; DmaAuth proposes a complementary hardware-level defense.

IOMMU Protection Against I/O Attacks (2018)

- **Citation:** "IOMMU protection against I/O attacks: a vulnerability and a proof of concept." Journal of the Brazilian Computer Society, Springer, 2018.
- **URL:** <https://link.springer.com/article/10.1186/s13173-017-0066-7>
- **Summary:** Demonstrated a specific IOMMU vulnerability and PoC. Relevant to ME's DMA protection mechanisms.

Intel IOMMU for DMA Protection in UEFI

- **Citation:** Intel, "A Tour Beyond BIOS: Using IOMMU for DMA Protection in UEFI Firmware." Intel White Paper.
- **URL:** <https://www.intel.com/content/dam/develop/external/us/en/documents/intel-whitepaper-using-iommu-for-dma-protection-in-uefi.pdf>
- **Summary:** Intel's official guidance on using IOMMU for DMA protection in firmware. Addresses the DMA attack class that RingWatch's ME DMA Protection module defends against.

2.3 Coprocessor-Based Monitoring

Co-processor-based Behavior Monitoring (2017)

- **Citation:** Chevalier, R., Villatel, M., Plaquin, D. (Hewlett-Packard). "Co-processor-based Behavior Monitoring." arXiv:1803.02700, December 2017.
- **URL:** <https://arxiv.org/abs/1803.02700>

- **Summary:** HP researchers proposed using a coprocessor (specifically the Intel ME) for behavior monitoring of the main system. The coprocessor runs independently and cannot be compromised by OS-level attacks (assuming the ME itself is not compromised).
- **Relevance to RingWatch:** This is the academic foundation for using the ME as a security monitoring platform — the dual-use nature of the ME. RingWatch currently monitors FROM the OS; this paper proposes monitoring FROM the ME.

A Coprocessor-Based Introspection Framework Via Intel Management Engine

- **Citation:** ISIS, Vanderbilt University.
- **URL:** <https://isis.vanderbilt.edu/publications/coprocessor-based-introspection-framework-intel-management-engine>
- **Summary:** Academic proposal for using the ME as an introspection framework — the ME can observe the main system's state without being observed itself.

A Framework to Secure Peripherals at Runtime

- **Citation:** Zhang, F., Wang, H., Leach, K., Stavrou, A. "A Framework to Secure Peripherals at Runtime." LNCS 8712.
- **URL:** <https://www.eecis.udel.edu/~hnw/paper/iocheck.pdf>
- **Summary:** Runtime peripheral security framework. Relevant to ME's role as a peripheral controller and its ability to secure/monitor other peripherals.

2.4 UEFI/BIOS Rootkits

LoJax (ESET, 2018)

- **Citation:** ESET, "LoJax: First UEFI rootkit found in the wild, courtesy of the Sednit group." WeLiveSecurity, September 2018.
- **URL:** <https://www.welivesecurity.com/2018/09/27/lojax-first-uefi-rootkit-found-wild-courtesy-sednit-group/>
- **MITRE ATT&CK:** S0397
- **Summary:** First UEFI rootkit found in the wild, deployed by APT28 (Fancy Bear/Sednit). Used RWEverything to write to SPI flash. Targeted government organizations in the Balkans and Central Asia.
- **Significance:** Proved that UEFI firmware persistence is not just theoretical — nation-state actors are using it in the wild.

MoonBounce (Kaspersky, 2022)

- **Citation:** Kaspersky, "MoonBounce: the dark side of UEFI firmware." Securelist, 2022.
- **URL:** <https://securelist.com/moonbounce-the-dark-side-of-uefi-firmware/105468/>
- **Summary:** Sophisticated UEFI firmware implant discovered by Kaspersky. MoonBounce is a malicious UEFI module that persists in SPI flash and executes before the OS loads.

BlackLotus (Microsoft/NSA, 2023)

- **Citation:** Microsoft, "Guidance for investigating attacks using CVE-2022-21894: The BlackLotus campaign." April 2023. NSA, "BlackLotus Mitigation Guide," June 2023.
- **URLs:**
 - <https://www.microsoft.com/en-us/security/blog/2023/04/11/guidance-for-investigating-attacks-using-cve-2022-21894-the-blacklotus-campaign/>
 - https://media.defense.gov/2023/Jun/22/2003245723/-1/-1/CSIBlackLotusMitigation_Guide.PDF
- **Summary:** BlackLotus is the first publicly known UEFI bootkit capable of bypassing Secure Boot on patched Windows systems. Exploits CVE-2022-21894 (BitLocker bypass). NSA published mitigation guidance.

2.5 Hardware Security & Side Channels

ROPAD (DAC 2020)

- **Citation:** "ROPAD: A Fully Digital Highly Predictive Ring Oscillator Probing Attempt Detector." DAC 2020.
- **URL:** <https://doi.org/10.1109/dac18072.2020.9218546>
- **Summary:** Hardware-level probing detector for microprobing attacks. Relevant to physical Ring -3 attacks via JTAG.

Programmable RO (2021)

- **Citation:** Yao, Y., Kiaei, P., Singh, R., Tajik, S., Schaumont, P. "Programmable RO (PRO): A Multipurpose Countermeasure against Side-channel and Fault Injection Attacks." IACR ePrint 2021/878.
- **URL:** <https://eprint.iacr.org/2021/878>
- **Summary:** Multipurpose countermeasure against side-channel and fault injection attacks. Relevant to ME side-channel defenses.

ShieldMMU (2025)

- **Citation:** Liu, G., Li, N., Chen, C. "ShieldMMU: Detecting and Defending against Controlled-Channel Attacks in Shielding Memory System." arXiv:2509.03879, September 2025.
- **URL:** <https://arxiv.org/abs/2509.03879>
- **Summary:** Memory system shielding against controlled-channel attacks. Relevant to ME's memory access patterns.

Interstellar (KAIST)

- **Citation:** "Interstellar: Fully Partitioned and Efficient Security Monitoring Hardware Near a Processor Core for Protecting Systems against Attacks on Privileged Software." KAIST.
- **URL:** <https://pure.kaist.ac.kr/en/publications/interstellar-fully-partitioned-and-efficient-security-monitoring-/>
- **Summary:** Hardware-based security monitoring near the processor core. Conceptually similar to RingWatch but implemented in hardware rather than software.

3. Industry Research & Threat Intelligence

3.1 Positive Technologies

Positive Technologies (PT) is the most prolific researcher of Intel ME vulnerabilities:

Research	Date	CVE	Significance
SA-00086: ME JTAG via USB	Nov 2017	CVE-2017-5705/5712	First ME JTAG activation — full system compromise
Where there's a JTAG, there's a way	Oct 2017	—	Detailed JTAG exploitation methodology
Debugging mechanism in Intel CPUs	Jan 2017	—	Intel CPU debug interface via USB 3.0
SA-00213: CSME IOMMU	May 2019	CVE-2019-0090	ME IOMMU bypass — DMA to host memory
Intel x86 Root of Trust: loss of trust	Mar 2020	—	Unfixable boot ROM vulnerability — cannot be patched
SA-00241: Red Unlock	2020	CVE-2020-8705	ME JTAG via physical resistor on PCH

3.2 Eclipsium

Eclipsium is the leading industry firmware security company:

Research	Date	Significance
Firmware and Frameworks — MITRE ATT&CK Mapping	2024	First systematic mapping of firmware attacks to MITRE ATT&CK
TrickBot Now Offers TrickBoot	2020	TrickBot gains UEFI/BIOS firmware inspection capability
Conti Opens a New Front in the Fight for Firmware	2022	Conti ransomware firmware attack capabilities
Firmware Security Realizations — Part 2: Start Your Management Engine	—	ME-specific firmware security analysis
Firmware Security Realizations — Part 3: SPI Write Protections	—	SPI flash write protection mechanisms
Enhanced Threat Detection: Bootloaders, Bootkits, and Secure Boot	—	Secure Boot bypass detection
UEFI Secure Boot Bypass in Dell Edge Gateway	—	Living-off-the-land Secure Boot bypass
Protecting System Firmware Storage	—	Firmware storage protection strategies

3.3 Intel Official Advisories & White Papers

Advisory	Date	CVEs	Scope
SA-00086	Nov 2017	CVE-2017-5705, CVE-2017-5712	ME 6.x-11.x, SPS 4.0, TXE 1.0-3.0
SA-00141	May 2018	—	AMT 9.x-12.x
SA-00213	May 2019	CVE-2019-0090	CSME IOMMU
SA-00241	2020	CVE-2020-8705	Red Unlock / ME JTAG
SA-01280	2025	—	2025.3 IPU chipset firmware
CVE-2023-40067	Aug 2024	CVE-2023-40067	CSME unchecked return value
CVE-2025-20037	2025	CVE-2025-20037	CSME privilege escalation
CVE-2025-27708	Feb 2026	CVE-2025-27708	CSME out-of-bounds read

3.4 Government/NIST Guidance

Document	Source	Date
NIST SP 800-147: BIOS Protection Guidelines	NIST	2011
BlackLotus Mitigation Guide	NSA	June 2023
Risk Assessment: Intel ME / vPro Hardware	Independent	April 2026

3.5 Reputable Red Team Forums & Community Tools

GitHub Repositories

Repository	Stars	Description
corna/me_cleaner	4,973	Partial deblobbing of Intel ME/TXE firmware images — primary ME neutralization tool
ptresearch/IntelTXE-PoC	545	Intel ME JTAG PoC (SA-00086) by Positive Technologies
peterbjornx/mesa86exploit	33	Exploit generator for Intel ME 11 buffer overflow
Jatinkapilaq1/intel-me-research	10	Zero-dependency Python HECI spy — direct ME communication
flothrone/intelme	16	Black Hat 2017 AMT research PDFs and demo videos
MangoKiwiPlumGrape/mecleanerthinkpad	13	me_cleaner fork with HAP bit fixes for 8th-14th gen Intel
yasindce1998/Barzakh	4	Ring -4 to Ring 0 offense-defense firmware research platform
patapik/uefi-persistence-whitepaper	0	UEFI firmware persistence whitepaper — attack techniques, detection rules, mitigations
osresearch/heads-wiki	—	LinuxBoot/Heads wiki — includes ME firmware cleaning guides

Community Research Blogs

Blog	Author	Content
KaKaRoTo's Blog	Younes (KaKaRoTo)	Multi-part series on reproducing Intel ME exploitation (SA-00086)
pbx.sh	Peter Bosch (@peterbjornx)	Intel ME deep dive talk, CVE-2019-11098 Boot Guard bypass
Purism Blog	Purism	Reverse-engineering the Intel ME's ROMP module
Positive Technologies Analytics	PT Research	Regular ME vulnerability research publications

4. Defensive Research Landscape

4.1 ME Neutralization (Offensive-Defensive)

me_cleaner (Nicola Corna, 4,973 stars) is the primary tool for ME neutralization:

- Sets the HAP bit (High Assurance Platform) in PCH STRP registers, which tells the ME to enter a minimal operational mode
- Removes non-essential ME modules from the firmware image
- Supported on ME 6.x through 14.x (with the ThinkPad fork extending to 18.x)
- Used by privacy-focused projects like Purism (Librem laptops), System76, and LinuxBoot/Heads
- **Risk:** Can brick systems if applied incorrectly — ME firmware must remain bootable for the system to POST

4.2 Boot Integrity Measurement

NIST SP 800-147 provides BIOS protection guidelines:

- Authenticated BIOS updates
- BIOS integrity measurement
- BIOS configuration protection
- However, NIST SP 800-147 predates widespread Ring -3 awareness and does not address ME-level attacks

4.3 Runtime Firmware Monitoring

Peacock (arXiv:2601.07402, January 2026) is the most recent academic work on runtime firmware monitoring:

- Proposes a UEFI firmware runtime observability layer
- Detects firmware modifications during runtime, not just at boot
- Relevant to RingWatch's runtime ME monitoring approach

4.4 DMA Protection

Intel IOMMU White Paper provides official guidance on using IOMMU for DMA protection:

- Inteliommu=on iommu=pt inteliommu=strict kernel parameters
- IOMMU groups for device isolation
- ACS (Access Control Services) for P2P DMA prevention (not available on consumer Meteor Lake-P)

RingWatch Implementation:

- IOMMU group monitoring (identity vs DMA-FQ lazy mode)
- Internal WiFi (wlo1) disabled — NC-SI isolation

- USB WiFi adapter used instead — traffic off ME's network path
- ACS limitation documented (consumer chip limitation)

4.5 HECI Bus Monitoring

RingWatch is the **first open-source tool** to provide real-time HECI bus monitoring:

- MEI client enumeration and state tracking
- HECI buffer monitoring (txqueuelimit, HBM version)
- ME firmware status register decoding
- SPI flash access monitoring via SPI Proxy client
- Network port monitoring (AMT ports 623, 664, 16990-16996)
- Power consumption tracking (RAPL)

5. Research Gaps & Future Directions

5.1 Unaddressed Areas

1. **No academic paper specifically addresses Ring -3 detection from the OS side** — RingWatch is novel in this regard
2. **No open-source ME firmware integrity validation tool exists** (beyond version string checking)
3. **No published research on NC-SI traffic detection** — RingWatch's NC-SI isolation (USB WiFi approach) is undocumented in academia
4. **No systematic mapping of MEI clients to attack surfaces** — RingWatch's client classification (POISON/PROTECT/MONITOR) is novel
5. **No academic treatment of RAM poisoning as anti-forensic defense** — Rutkowska (Black Hat DC 2007) is the closest, but it predates ME-specific research

5.2 Future Research Directions

1. **ME firmware diff analysis** — comparing ME firmware versions to detect malicious modifications
2. **HECI protocol fuzzing** — systematic fuzzing of MEI client interfaces to discover new vulnerabilities
3. **NC-SI traffic analysis** — network traffic capture and analysis on the NC-SI interface
4. **ME-based intrusion detection** — using the ME's out-of-band capability as a security monitor (per Chevalier et al. 2017)
5. **Cross-platform Ring -3 monitoring** — extending RingWatch to AMD PSP (Platform Security Processor) and ARM TrustZone

6. Bibliography Summary

Total sources catalogued:

- **Conference presentations:** 9 (Black Hat x3, DEF CON x1, CCC x2, GSEC/HITB x1, SICHERHEIT x1, EuroS&P x1)
- **Academic papers:** 12 (arXiv x5, USENIX x1, NDSS x1, Springer x2, IACR x1, LNCS x1, scitepress x1)
- **Industry research reports:** 15 (Positive Technologies x6, Eclysium x8, ESET x1)
- **Government/standards:** 3 (NIST x1, NSA x1, Independent x1)
- **Official advisories:** 8 (Intel SA-00086 through SA-01280 + standalone CVEs)
- **Community tools:** 9 (GitHub repositories)
- **Community blogs:** 4

Total: 60 sources

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